

EDUSAMBAM ASSESSMENT SERIES

Practice Test: Vectors & Transformations

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Subject	Mathematics (Maths)	Topic	Practice Test: Vectors & Transformations	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. A vector has:

- A. Only magnitude
- C. Both magnitude and direction

- B. Only direction
- D. Neither

Answer: ☐ A ☐ B ☐ C ☐ D

Basic

2. $a = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, $b = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$. Find $a + b$.

- A. $\begin{pmatrix} 5 \\ 4 \end{pmatrix}$
- C. $\begin{pmatrix} -1 \\ 6 \end{pmatrix}$

- B. $\begin{pmatrix} 5 \\ 6 \end{pmatrix}$
- D. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$

Answer: ☐ A ☐ B ☐ C ☐ D

Basic

3. $a = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$, $b = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$. Find $a - b$.

- A. $\begin{pmatrix} 8 \\ 11 \end{pmatrix}$
- C. $\begin{pmatrix} 4 \\ 11 \end{pmatrix}$

- B. $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$
- D. $\begin{pmatrix} 4 \\ -3 \end{pmatrix}$

Answer: ☐ A ☐ B ☐ C ☐ D

Intermediate

4. $v = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$. Find $3v$.

- A. $\begin{pmatrix} 6 \\ -5 \end{pmatrix}$
- C. $\begin{pmatrix} 3 \\ -5 \end{pmatrix}$

- B. $\begin{pmatrix} 9 \\ -15 \end{pmatrix}$
- D. $\begin{pmatrix} 9 \\ 15 \end{pmatrix}$

Answer: ☐ A ☐ B ☐ C ☐ D

Basic

5. The position vector of point A relative to origin O is written:

- A. $\rightarrow AO$
- C. A^2

- B. $|OA|$
- D. $\rightarrow OA$

Answer: ☐ A ☐ B ☐ C ☐ D

Basic

6. A has position vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, B has position vector $\begin{pmatrix} 7 \\ 6 \end{pmatrix}$. Find $\rightarrow AB$.

- A. $\begin{pmatrix} 8 \\ 8 \end{pmatrix}$
- C. $\begin{pmatrix} -6 \\ -4 \end{pmatrix}$

- B. $\begin{pmatrix} 6 \\ 4 \end{pmatrix}$
- D. $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$

Answer: ☐ A ☐ B ☐ C ☐ D

Intermediate

7. Find the magnitude of $v = \begin{pmatrix} 8 \\ 15 \end{pmatrix}$.

- A. 17
- C. 7

- B. 23
- D. 120

Answer: ☐ A ☐ B ☐ C ☐ D

Intermediate

8. A full description of a reflection must include:

- A. A scale factor
- C. The equation of the mirror line

- B. A centre point
- D. A direction

Answer: ☐ A ☐ B ☐ C ☐ D

Basic

9. Point B(5, 2) is reflected in the x-axis. Find B'.

- A. (-5, 2)
- C. (2, 5)

- B. (5, -2)
- D. (-5, -2)

Answer: ☐ A ☐ B ☐ C ☐ D

Intermediate

10. Point C(6, 1) is reflected in the line $y = x$. Find C'.

- A. (6, -1)
- C. (-1, -6)

- B. (-6, 1)
- D. (1, 6)

Answer: ☐ A ☐ B ☐ C ☐ D

Intermediate

11. A full description of a rotation must include:

- A. Only the angle
- C. Centre, angle, AND direction

- B. Only the centre
- D. Only the direction

Answer: ☐ A ☐ B ☐ C ☐ D

Intermediate

12. Point D(2, 7) is rotated 90° anticlockwise about the origin, using $(x, y) \rightarrow (-y, x)$. Find D'.

Advanced

- A. (-7, 2)
C. (7, 2)

- B. (2, -7)
D. (7, -2)

Answer: ☐ A ☐ B ☐ C ☐ D

13. A full description of an enlargement must include:

Intermediate

- A. Centre AND scale factor
C. A direction only

- B. A mirror line
D. An angle only

Answer: ☐ A ☐ B ☐ C ☐ D

14. Point E(3, 5) is enlarged by scale factor 2, centre the origin. Find E'.

Intermediate

- A. (5, 7)
C. (6, 10)

- B. (1.5, 2.5)
D. (9, 15)

Answer: ☐ A ☐ B ☐ C ☐ D

15. A negative scale factor produces an image that is:

Advanced

- A. On the same side, right-way up
C. Always larger, never smaller

- B. On the opposite side of the centre, and upside down
D. The same size and position as the object

Answer: ☐ A ☐ B ☐ C ☐ D

Section B — True / False

20 marks

16. Two vectors are equal if they have the same magnitude and direction.

Basic

☐ True ☐ False

17. If $\mathbf{a}$ and $\mathbf{b}$ are the position vectors of A and B, then $\rightarrow\mathbf{AB} = \mathbf{a} - \mathbf{b}$.

Intermediate

☐ True ☐ False

18. Two vectors are parallel if one is a scalar multiple of the other.

Intermediate

☐ True ☐ False

19. The magnitude of a column vector (x, y) is found using $\sqrt{x^2 + y^2}$.

Basic

☐ True ☐ False

20. The position vector of the midpoint M of AB is $\mathbf{a} + \mathbf{b}$ (not halved).

Intermediate

☐ True ☐ False

21. A translation is fully described by a column vector.

Basic

☐ True ☐ False

22. A rotation can be fully described using only the angle, with no centre needed.

Intermediate

☐ True ☐ False

23. Under a reflection, every point on the mirror line is invariant.

Intermediate

☐ True ☐ False

24. A negative scale factor always makes the image larger, never smaller.

Advanced

☐ True ☐ False

25. Two translations applied one after another are equivalent to a single translation using the sum of the two vectors.

Intermediate

☐ True ☐ False

Section C — Fill in the Blank

20 marks

26. A quantity with both magnitude and direction is called a _____.

Answer:

27. If $\mathbf{a}$ and $\mathbf{b}$ are the position vectors of A and B, then $\vec{AB} = \mathbf{b} - \underline{\hspace{2cm}}$.

Answer:

28. The magnitude of vector (x, y) is $\sqrt{x^2 + \underline{\hspace{2cm}}}$.

Answer:

29. Two vectors are $\underline{\hspace{2cm}}$ if one is a scalar multiple of the other.

Answer:

30. The position vector of the midpoint of AB is $\underline{\hspace{2cm}}$ $(\mathbf{a} + \mathbf{b})$.

Answer:

31. A translation is described fully using a column $\underline{\hspace{2cm}}$.

Answer:

32. A rotation must be described using a centre, an angle, and a $\underline{\hspace{2cm}}$.

Answer:

33. An enlargement is described using a centre and a $\underline{\hspace{2cm}}$ factor.

Answer:

34. A negative scale factor produces an image that is $\underline{\hspace{2cm}}$ down relative to the object.

Answer:

35. The invariant point of an enlargement is the $\underline{\hspace{2cm}}$ of enlargement.

Answer:

Section D — Column Matching

10 marks

36–40. Match each term to its correct definition.

Column A	Ans.	Column B
Translation	$\underline{\hspace{2cm}}$	A. One vector is a scalar multiple of the other
Reflection	$\underline{\hspace{2cm}}$	B. Described fully by a column vector
Rotation	$\underline{\hspace{2cm}}$	C. Described fully by a centre and a scale factor
Enlargement	$\underline{\hspace{2cm}}$	D. Described fully by the equation of the mirror line
Parallel vectors	$\underline{\hspace{2cm}}$	E. Described fully by a centre, an angle, and a direction

Section E — Short Answer

20 marks

41. Explain how to find the vector from point A to point B, given their position vectors.

42. Describe how to find the midpoint of a line segment using position vectors.

43. Explain why a full description of a rotation must include the direction (clockwise or anticlockwise).

44. Describe what happens to a shape when it is enlarged by a negative scale factor.

45. Explain what an "invariant point" is, and give one example from a specific transformation.

Candidate Signature: _____

Invigilator Signature: _____

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Marks Obtained: _____ / 100 Checked By: _____

Result / Grade: _____ Date Checked: _____

Practice Test: Vectors & Transformations — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. C | |
| 2. A | 9. B |
| 3. D | 10. D |
| 4. B | 11. C |
| 5. D | 12. A |
| 6. B | 13. A |
| 7. A | 14. C |
| 8. C | 15. B |

Section B — True / False

- | | |
|-----------|-----------|
| 16. True | 21. True |
| 17. False | 22. False |
| 18. True | 23. True |
| 19. True | 24. False |
| 20. False | 25. True |

Section C — Fill in the Blank

- | | |
|--------------|---------------|
| 26. vector | 31. vector |
| 27. a | 32. direction |
| 28. y^2 | 33. scale |
| 29. parallel | 34. upside |
| 30. $1/2$ | 35. centre |

Section D — Column Matching

- | | |
|---------------------|--------------------------|
| 36. Translation — B | |
| 37. Reflection — D | 39. Enlargement — C |
| 38. Rotation — E | 40. Parallel vectors — A |

Section E — Short Answer (Model Answer Keywords)

41. $b-a$; subtract; position vectors
42. half; $a+b$; average
43. two directions; ambiguous; different image
44. opposite side; upside down; flipped
45. maps to itself; unchanged; mirror line; centre